

MODULE 5 OF 6

Grade Estimation: From Inverse Distance to Kriging

We estimate grade where we did not sample, and we derive the best possible way to do it

Everything so far has been preparation. We build from classical estimators, derive the ordinary kriging system in full from its two defining requirements, and run it on the Toro data — honestly reporting what 26 pits can and cannot deliver.

LEARNING OBJECTIVES

- Write any linear estimator as a weighted sum of samples.
- Apply and critique polygonal and inverse-distance estimators.
- Derive the ordinary kriging system from unbiasedness and minimum variance using a Lagrange multiplier.
- Interpret the kriging variance and the properties of kriging.
- Distinguish point and block kriging and know the main variants.
- Validate an estimate by cross-validation and swath analysis.

1 The estimation problem

We have grade at n sampled locations and we want grade at a location where we did not sample. Every practical method answers this as a **weighted linear combination** of the surrounding samples:

$$Z^*(\mathbf{x}_0) = \sum_{i=1}^n \lambda_i Z(\mathbf{x}_i) \quad (5.1)$$

Every estimator in this module differs only in how it sets the weights λ_i .

2 Classical estimators

Polygonal / nearest-neighbour: the nearest sample takes weight 1, all others 0. Transparent and honest with the data but a step function, with no measure of uncertainty.

Inverse distance weighting: weight each sample by the inverse of its distance, raised to a power p :

$$Z^*(\mathbf{x}_0) = \frac{\sum_{i=1}^n d_i^{-p} Z(\mathbf{x}_i)}{\sum_{i=1}^n d_i^{-p}}, \quad d_i = \|\mathbf{x}_i - \mathbf{x}_0\| \quad (5.2)$$

The power p is arbitrary; it uses only distance and is blind to the variogram; it ignores data clustering (three close samples out-vote one lone one); and it returns no uncertainty. Kriging was invented to fix all four.

3 The criteria for an optimal estimator

Kriging — named for Danie Krige by Matheron — demands two things of the estimate. Define the estimation error $R(\mathbf{x}_0) = Z^*(\mathbf{x}_0) - Z(\mathbf{x}_0)$.

Requirement 1 — unbiasedness: $\mathbb{E}[R] = 0$. The estimate neither over- nor under-states grade on average.

Requirement 2 — minimum variance: among all estimators that satisfy requirement 1, choose the one whose error has the smallest variance.

Kriging is the BLUE

An estimator that is a linear combination of the data, unbiased, and with minimum achievable error variance, is the **Best Linear Unbiased Estimator**. Kriging *is* the BLUE for a regionalised variable.

4 Deriving the ordinary kriging system

Ordinary kriging is used when the mean m is constant but *unknown*.

Step 1 — unbiasedness

With $\mathbb{E}[Z(\mathbf{x}_i)] = m$ for all i :

$$\mathbb{E}[R] = m \sum_i \lambda_i - m = m \left(\sum_i \lambda_i - 1 \right)$$

For this to vanish for unknown m :

$$\sum_{i=1}^n \lambda_i = 1 \quad (5.4)$$

Step 2 — the estimation variance

$$\sigma_E^2 = \text{Var}[R] = \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j C(\mathbf{x}_i, \mathbf{x}_j) - 2 \sum_{i=1}^n \lambda_i C(\mathbf{x}_i, \mathbf{x}_0) + C(\mathbf{x}_0, \mathbf{x}_0) \quad (5.5)$$

The first term punishes redundancy among the samples; the middle term rewards proximity to the target. Kriging will automatically reward proximity and penalise clustering — something IDW could never do.

Step 3 — constrained minimisation by Lagrange multiplier

Form the Lagrangian (the factor of 2 is cosmetic):

$$L(\lambda_1, \dots, \lambda_n, \mu) = \sigma_E^2 - 2\mu \left(\sum_{i=1}^n \lambda_i - 1 \right) \quad (5.6)$$

At the minimum every partial derivative is zero:

$$\frac{\partial L}{\partial \lambda_i} = 2 \sum_{j=1}^n \lambda_j C(\mathbf{x}_i, \mathbf{x}_j) - 2C(\mathbf{x}_i, \mathbf{x}_0) - 2\mu = 0$$

and $\partial L / \partial \mu = 0$ returns the constraint (5.4). Dividing by 2 gives:

The ordinary kriging system (covariance form)

$$\sum_{j=1}^n \lambda_j C(\mathbf{x}_i, \mathbf{x}_j) - \mu = C(\mathbf{x}_i, \mathbf{x}_0), \quad i = 1, \dots, n$$

$$\sum_{j=1}^n \lambda_j = 1$$

$n + 1$ linear equations in $n + 1$ unknowns: the n weights and the multiplier μ .

Using $C(\mathbf{h}) = C(0) - \gamma(\mathbf{h})$ the system is rewritten in the **variogram** (the form we actually modelled in Module 4):

$$\sum_{j=1}^n \lambda_j \gamma(\mathbf{x}_i, \mathbf{x}_j) + \mu = \gamma(\mathbf{x}_i, \mathbf{x}_0), \quad \sum_{j=1}^n \lambda_j = 1 \quad (5.7)$$

Step 4 — the kriging variance

Substituting the solved weights into (5.5) collapses to:

$$\sigma_{OK}^2 = \sum_{i=1}^n \lambda_i \gamma(\mathbf{x}_i, \mathbf{x}_0) + \mu \quad (5.8)$$

This is what separates kriging from every classical method: it delivers, for free, a quantified uncertainty alongside each estimate — the third clause of Module 1's central question, finally answered with mathematics.

5 The kriging system in matrix form

$$\begin{bmatrix} \Gamma_{11} & \cdots & \Gamma_{1n} & 1 \\ \vdots & \ddots & \vdots & \vdots \\ \Gamma_{n1} & \cdots & \Gamma_{nn} & 1 \\ 1 & \cdots & 1 & 0 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_n \\ \mu \end{bmatrix} = \begin{bmatrix} \gamma_{10} \\ \vdots \\ \gamma_{n0} \\ 1 \end{bmatrix} \quad (5.9)$$

where $\Gamma_{ij} = \gamma(\mathbf{x}_i, \mathbf{x}_j)$ and $\gamma_{i0} = \gamma(\mathbf{x}_i, \mathbf{x}_0)$. The left matrix depends only on data geometry, so it is built once; only the right vector changes per target.

```
import numpy as np
from scipy.spatial.distance import cdist

def build_kriging_matrix(xc, yc, gamma_model):
    """Assemble matrix A of equation (5.9) - built once for all targets."""
    n = len(xc)
    A = np.zeros((n + 1, n + 1))
    A[:n, :n] = gamma_model(cdist(np.c_[xc, yc], np.c_[xc, yc]))
    A[:n, n] = 1.0          # constraint column
    A[n, :n] = 1.0          # constraint row
    A[n, n] = 0.0
    return A

def ok_estimate(x0, y0, xc, yc, zc, Ainv, gamma_model):
    """Estimate and kriging variance at one target point."""
    n = len(zc)
    d0 = np.sqrt((xc - x0)**2 + (yc - y0)**2)
    b = np.empty(n + 1)
    b[:n] = gamma_model(d0)    # sample-to-target variogram
    b[n] = 1.0                # constraint
    w = Ainv @ b               # lambda_1..n and mu
    estimate = float(np.dot(w[:n], zc))    # equation (5.1)
    variance = float(np.dot(w[:n], b[:n]) + w[n])    # equation (5.8)
    return max(estimate, 0.0), max(variance, 0.0)
```

6 Properties of kriging

Exact interpolator. With no nugget, kriging at a sampled location returns that sample's grade exactly and a kriging variance of zero. Add a nugget and exactness is broken — kriging smooths through data points because the nugget tells it the data themselves carry noise.

Smoothing. $\text{Var}[Z^*] < \text{Var}[Z]$ always. High grades are estimated too low, low grades too high, and the block-model histogram is narrower than reality.

Because a kriged model compresses the grade distribution towards the mean, selecting ore at a high cut-off tends to discover real grade lower than the model promised — **conditional bias**. It is why neighbourhood design (QKNA) and conditional simulation (Module 6) exist.

Automatic declustering. The sample-to-sample matrix Γ_{ij} in (5.9) makes clustered samples share weight.

Screen effect and negative weights. A sample between target and a distant sample screens the distant one, sometimes cutting its weight below zero. Useful but can give nonsensical grades; often constrained.

Uses the variogram, hence the geology. Unlike IDW, weights depend on the nugget, sill, range and anisotropy measured in Module 4.

Neighbourhood is a design choice. A search radius near the range, minimum and maximum sample counts, and often octant searches: designed by **Quantitative Kriging Neighbourhood**

Analysis (QKNA).

7 Point and block kriging

A mine extracts **blocks**, not points. **Block kriging** replaces the sample-to-target variogram on the right of (5.7) with the average sample-to-block variogram:

$$\bar{\gamma}(\mathbf{x}_i, V) = \frac{1}{|V|} \int_V \gamma(\mathbf{x}_i, \mathbf{u}) d\mathbf{u} \approx \frac{1}{N} \sum_{k=1}^N \gamma(\mathbf{x}_i, \mathbf{u}_k) \quad (5.10)$$

Because a block average is intrinsically less variable than a point, block kriging yields a smaller kriging variance. Block kriging is the standard for resource block models.

8 The kriging family

Simple kriging assumes a known mean; weights need not sum to one. **Lognormal kriging** estimates in log space with a rigorous back-transform. **Indicator kriging** is non-parametric and rebuilds the local grade distribution from indicator thresholds — strong for highly skewed data. **Universal kriging** handles a systematic spatial trend. **Co-kriging** uses a correlated secondary variable; at Toro Nb₂O₅ correlates with SnO₂ at $r = 0.76$ — a genuine co-kriging candidate.

9 Validating the estimate

Cross-validation is the central test: remove one sample, kriging its location from the remaining $n - 1$, compare with the withheld value; repeat for every sample. Three numbers: *mean error* (tests bias), *RMSE* (tests precision), *slope of regression* of true on estimate (tests conditional bias).

Swath plots divide the deposit into slices and compare slice means of estimates and data; they expose smoothing and local bias. **Visual validation** posts drill data on the model. **Global statistics** check that the kriged mean reproduces the declustered data mean.

10 The Toro estimate

The full machinery runs on the Toro composited pits with the Module 4 spherical model. The kriging matrix is built once; the system is solved on a 60 m grid.

Metric	Value	Reading
Mean error	+0.02% SnO ₂	essentially unbiased — good
RMSE	1.80% SnO ₂	large against a 2.3% mean grade
Correlation est–actual	0.24	weak predictive power

The kriging is *unbiased* — requirement 1 satisfied. But its *precision is poor*: correlation 0.24 means the estimate barely out-predicts the global mean. That is not a failure of kriging — the BLUE derivation guarantees these are the best linear weights obtainable. It is a failure of *data*. With a relative nugget of 0.55 and 26 widely spread pits, there is too little spatial

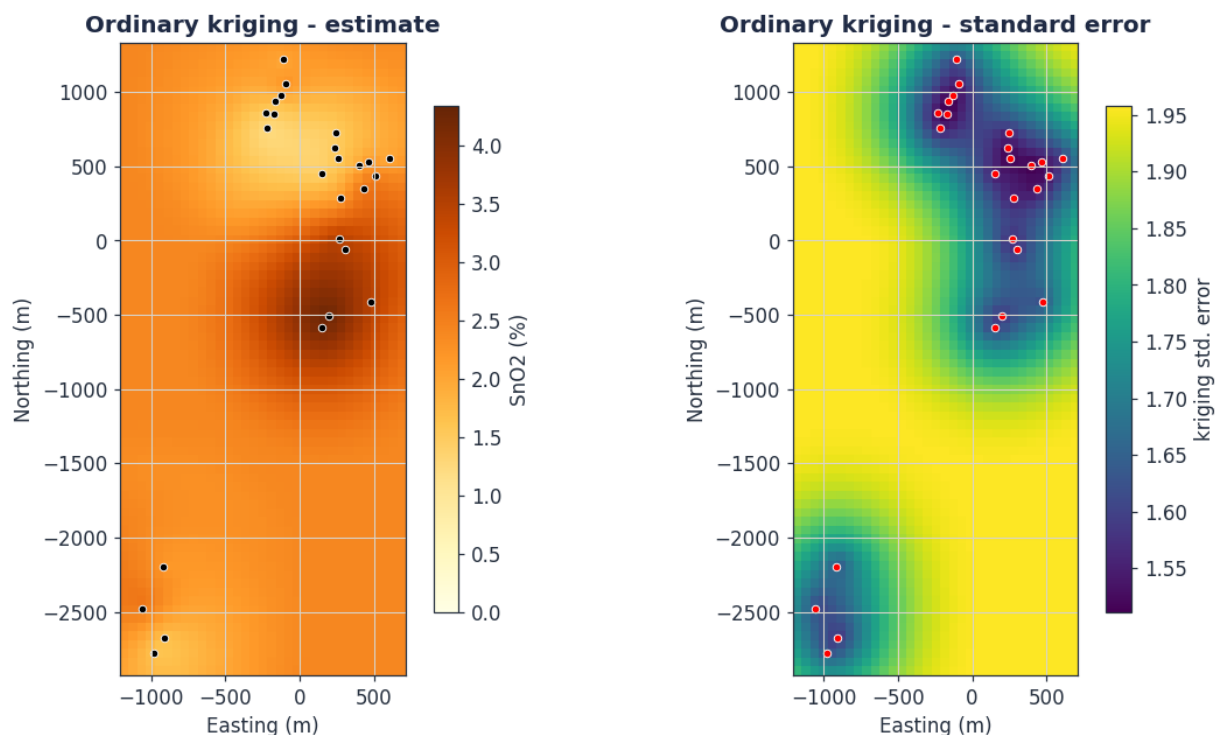


Figure 1 – Toro ordinary kriging. Left: grade estimate. Right: kriging standard error from equation (5.8). The standard-error map — which no classical estimator can draw — says exactly where the estimate can be trusted and drives the classification of Module 6.

signal. The mathematics has done its job; it has also told you the truth.

Why this is the most important result in the course

A weak cross-validation is not a setback to be hidden — it is the estimate quantifying confidence honestly. It is the mathematical proof that Toro is an Exploration Target. And it is *prescriptive*: precision will improve only by attacking the nugget (better sampling protocols, Module 2) and adding data density (drilling). Kriging has not just estimated the deposit; it has costed and justified the next exploration budget.

Module 5 of the Toro Resource Estimation course. Next: Module 6, Block Models, Cut-off Grade and the Resource Statement.